OXFORD INTERNATIONAL AQA EXAMINATIONS

INTERNATIONAL GCSE
Biology
9201/1

Paper 1

Mark Scheme

November 2018

Version: 1.0 Final

18BY92011/MS

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Copyright © 2018 Oxford International AQA Examinations and its licensors. All rights reserved.

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

<table>
<tr><th>Question</th><th>Answers</th><th>Extra information</th><th>Mark</th><th>AO / Spec. Ref.</th></tr>
<tr><td>01.1</td><td>A. cell membrane
B. nucleus</td><td></td><td>1
1</td><td>AO1
3.1.1a</td></tr>
<tr><td>01.2</td><td>$\frac{12.5}{2500}$

0.005 (mm)</td><td>an answer of 0.005 mm scores 2 marks

allow 5×10^{-3}</td><td>1

1</td><td>AO2
3.1.1a
6.3.10</td></tr>
<tr><td>01.3</td><td>to release energy

(for) movement (of tail)</td><td>do not allow produce / create / make energy</td><td>1

1</td><td>AO1/2
3.1.1a/b
3.2.6b/e/f</td></tr>
<tr><td>01.4</td><td><table>
<tr><th rowspan="2">Characteristic</th><th colspan="3">Characteristics due to</th></tr>
<tr><th>Environment only</th><th>Genes only</th><th>Genes and the environment</th></tr>
<tr><td>Eye Colour</td><td></td><td>✓</td><td></td></tr>
<tr><td>Height</td><td></td><td></td><td>✓</td></tr>
<tr><td>Scar on leg</td><td>✓</td><td></td><td></td></tr>
<tr><td>Skin colour</td><td></td><td></td><td>✓</td></tr>
</table></td><td>allow 1 mark for 2 or 3 correct</td><td>2</td><td>AO2
3.5.3a</td></tr>
<tr><td>01.5</td><td>(one X and) one Y chromosome</td><td></td><td>1</td><td>AO2
3.5.3c/d
3.5.2c</td></tr>
<tr><td>01.6</td><td>23</td><td></td><td>1</td><td>AO1
3.5.2g/h</td></tr>
<tr><td>01.7</td><td>mitosis

copies of genetic material made

(two) genetically identical [body] cells</td><td>

allow genes / chromosomes / DNA is doubled / duplicated</td><td>1

1

1</td><td>AO1
3.5.2e</td></tr>
<tr><td>Total</td><td></td><td></td><td>13</td><td></td></tr>
</table>

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	an enzyme a vector an early	must be in correct order	1 1 1	AO1 3.5.5b/c

Question	Answers	Mark	AO / Spec. Ref.
02.2	**Level 3:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5-6	AO2 3.5.5e/f
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3-4	AO3 3.5.5d
	Level 1: Relevant points are made. They are not logically linked.	1-2	AO3 3.5.5f
	No relevant content	0	
	Indicative content allow converse arguments **Advantages of golden rice** • rice is a staple food in many developing countries (where vit A deficiency is most common) • reduces risk of blindness • reduces serious risks of infectious diseases (due to improved immunity) • (so) fewer deaths (from vit A deficiency) **Disadvantages of golden rice** • expensive to make golden rice • possible increase in malnutrition (as fewer vegetables consumed) • requires preparation / cooking before eating **Justifications** • tablets cheaper **but** may be harder to organise a programme of delivery • rice available all year round **but** fruit/veg may be seasonal • golden rice may cost more **but** buying sufficient fruit / veg may be more expensive (out of season) **Additional detailed knowledge** • golden rice may generally have increased yields (compared to normal rice) • possible effects on wild flowers and insects • unknown (long term) effects of eating GM crops on human health		

Total			**9**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	any **two** from: • cell wall • (permanent) vacuole • chloroplasts		2	AO1 3.1.1a
03.2	ribosomes		1	AO1 3.1.1a
03.3	nucleus		1	AO1 3.1.1a
03.4	section of DNA [that] codes for a protein		1 1	AO1 3.5.3i
03.5	weak/thin blood vessel wall/lining likely to burst/leak	 allow rupture for burst/leak	1 1	AO3 3.2.3j
03.6 View with Figure 6 Mark with 3.7	identification of change from CGC to CCC	allow if clearly identified on Figure 6	1	AO2 3.5.3k
03.7 Mark with 3.6	(amino acid changed from) arginine to proline	allow ecf from 03.6	1	AO2 3.5.3k

03.8	genetic diagram including: correct gametes **or** parental genotypes **D** and **d** <u>and</u> **d** and **d** **or** **Dd** <u>and</u> **dd** offspring genotypes correctly <u>derived:</u> **Dd Dd dd dd** identification of **Dd** as child with EDS 0.5 / ½ / 1 in 2 / 50% / 1 : 1	allow alternative symbols if defined do **not** allow if alternative symbols used and **not** defined allow genotypes correct for student's parental gametes allow correct identification of student's offspring genotypes allow ecf from student's derivation.	 1 1 1 1	AO2/3 3.5.3g 3.5.4a

Total			**14**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	1. Light receptors – eyes 2. Sound receptors – ears		1 1	AO1 3.4.2c
04.2	any **four** from: • (receptor to) sensory neurone • (ref to) CNS **or** spinal cord **or** relay neurone • (ref to) motor neurone • (ref to) muscles • (ref to) synapse (at any suitable point)	Full marks can only be given if sequence is in the correct order	4	AO1 3.4.1d/e
04.3	fewer receptors (so) fewer impulses sent along sensory neurones **or** (so) fewer impulses sent to the brain		1 1	AO2 3.4.1b 3.4.2b/c
04.4	both bars drawn to the correct height		1	AO2 3.2.4a 3.4.1b
04.5	capsaicin is insoluble (so) will not be dissolved / removed by water impulse still sent (from receptors)	allow very low solubility allow remains on tongue	1 1	AO3 3.2.4a 3.4.1b

04.6	starch is insoluble (so) the concentration (of solution) inside the cell stays the same (and therefore) water will not pass in by osmosis (and cause the cell to swell)	allow converse points re glucose	1 1 1	AO1/2 3.2.4a 3.1.5d/e

Total			**14**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$	allow one mark for correct symbol equation without balancing	2	AO1 3.2.1a
05.2	any **two** from • same species of snail • same species of pondweed • same size / mass of pondweed • number of snails • volume of indicator solution • temperature		2	AO4 3.2.6m
05.3		must be clear that water snails respire and pondweed photosynthesises allow ref to pondweed respiring (continuously)		AO1/2 3.2.1b 3.2.6c/d
	water snails respire		1	
	(and respiration) produces carbon dioxide		1	
	pondweed have light for photosynthesis		1	
	(and photosynthesis) requires carbon dioxide		1	
	(therefore) no (overall) change of carbon dioxide (so indicator stays red).		1	
05.4	yellow		1	AO2 3.2.1b 3.2.6c/d
	no light so no photosynthesis but respiration produces CO_2		1	

05.5	any **one** hazard from • allergies to snails **or** plants • risk of disease from snails any **one** precaution from • wear eye protection / goggles • wash hands • cover any cuts on exposed hands and arms • do not taste / eat plants • do not eat snails	precaution must match suitable hazard	1 1	AO4 3.2.6m

Total			**13**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	A: trachea B: rib	allow backbone / vertebrae	1 1	AO1 3.2.5a
06.2	diffusion from high concentration to low concentration		1 1	AO1 3.2.5.a/c 3.1.5a/b
06.3	$\frac{2.45-(-2.60)}{-2.60} \times 100$ 194.231 194 (%)	Usual allowance of ± half a small square means a reading of +2.4 or +2.5 is acceptable instead of +2.4 an answer of 194 scores 2 marks	1 1 1	AO2 3.2.5b/c 6.3.3 6.3.4 6.3.16
06.4	diaphragm contracts **or** flattens **or** intercostal muscle contracts (to raise ribs) which causes the volume (of the thorax) to increase causing pressure (in thorax) to decrease below zero	must be in the correct order allow bellow atmospheric pressure	1 1 1 1	AO2 AO3 3.2.5b

06.5	many filaments (gives large surface area) filaments thin **or** short diffusion pathway	allow ref to large surface area alone. **or** allow water flowing over gills (maintains concentration gradient) **or** good blood supply (maintains concentration gradient)	1 1	AO2 3.1.5i/j

06.6	lure/light lets the anglerfish see / attract prey to get food **or** lure/light lets the angler fish see / attract mates (1) to reproduce (1) (slow moving so) less energy needed **or** less respiration required (so) less ‘heat is lost **or** (so) energy conserved	 allow to avoid being eaten by predators	1 1 1 1	AO2 3.3.2a/d/e/f

Total			**17**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	microorganism that causes disease	allow bacteria / virus / fungi for microorganism	1	AO1 3.4.7a
07.2	dead / inactive pathogen (in vaccine) causes / stimulates white blood cells to produce antibodies many antibodies produced **<u>rapidly</u>** on reinfection	allow weakened **or** antigen from pathogen	1 1 1	AO1 3.4.7e
07.3	pathogen strains have different antigens the people who are immune to strain A have specific white cells / antibodies to strain A / antigen A which will **not** kill / destroy / bind to strain B.		1 1 1	AO2 AO3 3.4.7c/d 3.2.3r/s
07.4	**<u>Hh</u>** for sickle cell more likely to survive (from malaria) (more likely to) breed successfully (so) allele **or** h passed on to the future generation HH people are more likely to die (from malaria) So the Hh people will breed more	allow fewer red blood cells destroyed allow gives selective advantage allow ref to genes	1 1 1	AO1/3 3.2.3n 3.6.2b 3.5.4a 3.3.2f
Total			**10**	